

Track 51: Robotics Engineering

June 2026

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Tier 2 — Strong Demand | Departments: Mechatronics · Mechanical · ECE

Skill path: Robot Kinematics → Sensors → Actuators → Control → ROS2 → Simulation → Autonomous Navigation → Manipulation

Related: [Track 19 — Robotics & IoT](#) · [Track 23 — ROS2](#) · [Track 42 — Mechatronics](#)

Track Overview

Who this is for: Mechatronics, Mechanical, and ECE students targeting robotics engineer roles (distinct from general IoT track).

Prerequisites: Python/C++; basic mechanics; [Track 22 Arduino](#) helpful.

Tools: ROS2, Gazebo, RViz2, Python/C++, motor drivers, LIDAR/camera sensors, TurtleBot or custom kit.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Robotics awareness	Simulated mobile robot
1 Month	80–100	Junior robotics dev	Sensor-fusion robot project
3 Months	250–300	Internship-ready	Autonomous navigation capstone
6 Months	500–600	Robotics engineer entry	Hardware + sim portfolio

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Robot anatomy, DOF, kinematics intro	Robot specification doc
D3–4	Sensors: encoder, IMU, LIDAR, camera	Sensor integration lab
D5–6	Actuators: DC/servo/stepper, motor drivers	Motor control system
D7–8	ROS2 nodes, topics, Gazebo simulation	Simulated robot movement
D9–10	Capstone	Autonomous line-follow or obstacle avoid bot

1-Month Foundation

Week	Topics	Project
W1	Kinematics, forward/inverse intro, URDF	Custom robot model
W2	ROS2 middleware, publishers/subscribers	Sensor data pipeline
W3	SLAM intro, mapping, localization	Generated environment map
W4	Nav2 stack, path planning	Point-to-point autonomous robot

3-Month Job Ready

Month	Topics	Projects
M1	Mobile robotics, wheeled/legged intro	Mobile robot platform
M2	Manipulation intro, gripper control	Pick-and-place sim
M3	Integration capstone	Warehouse or delivery robot scenario

6-Month Professional

Add: real hardware deployment, computer vision ([Track 26](#)), multi-robot coordination, industry case study.

Assessment Rubric

Weekly Quiz 20% | Lab/simulation 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Tier 2 “Robotics” marketing course. IoT breadth → [Track 19](#). ROS2 depth → [Track 23](#). This track balances both for robotics job roles.